

Paper B-05: Neural Computation: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This paper treats neural computation as an engineered and biological transport analogy for CDFD/AFL. Spikes, synaptic current, activation flow, gradients, and attention-like routing are read as candidate fluxes Φ moving through constraints C such as bandwidth, inhibition, metabolic cost, finite precision, and connectivity. The paper does not claim that biological brains and artificial networks are the same material system. It asks which measurable bottlenecks and memory terms make neural computation auditable.

1 Series Context

Part I sets the flow-constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. This paper sits in Part B because it focuses on the engineered-computation reading of neural systems.

Notation. In this paper, C names the active computational constraint or capacity burden. Φ is the signal or update flow, S is adaptive response, and M_s is retained structural memory. Λ is used here only when referencing the Part II Life Number [2]; it is not used as a claim about consciousness or machine life.

2 Neural Computation as Transport

Neural computation moves signals through constrained pathways. In biological tissue, those constraints include synaptic strength, inhibition, refractory periods, metabolic cost, and network topology. In artificial neural networks, they include parameter count, activation bandwidth, context length, memory traffic, precision, optimization state, and hardware limits. CDFD/AFL gives a single audit question for both settings: where does flow concentrate, what limits it, how does the system adapt, and what past load remains in the structure?

3 Candidate Variable Map

CDFD term	Neural-computation reading	Example proxy
Φ	signal, activation, or gradient flow	spike rate, activation norm, token updates
C	active constraint	inhibition, bandwidth, precision, memory traffic
S	responsiveness	plasticity, gating, normalization, routing adjustment
M_s	structural memory	synaptic weights, trained parameters, cached state
Ψ_s	operating ratio	balanced, saturated, unstable, or underused regime

4 Biological and Engineered Boundaries

Part III gives the biological and medical context for AFL [3]. This paper uses that context carefully. A biological neuron, a neuromorphic device, and a transformer layer can all be described as constrained signal systems, but the measurements differ. A useful comparison must state the unit of flow, the time scale, the constraint, and the failure mode before drawing any cross-domain conclusion.

5 Measurement Layer

For biological data, measurable proxies may include firing rate, synaptic current, local field power, metabolic demand, refractory recovery, and plastic change. For artificial systems, proxies may include activation magnitude, gradient norm, attention entropy, memory bandwidth, context pressure, and inference latency. CDFD/AFL adds value only if these proxies improve the audit of saturation, collapse, recovery, or adaptation.

6 Current Runtime Diagnostic

The Part B adapter sweep includes artificial-intelligence and computation rows, and the Part E cascade test records finite runtime behavior in the current CDFD Runtime [4]. These files document model behavior. They do not validate a neuroscience claim or certify an AI-system claim.

7 Tests That Can Break the Mapping

The mapping weakens if measured bottlenecks do not track saturation or recovery, if ordinary neural-network or neuroscience metrics explain the behavior more cleanly, or if the proposed variables cannot be measured without circular definitions.

8 Conclusion

Neural computation is a useful Part B case because it sits between biological signal processing and engineered computation. The CDFD/AFL frame is strongest when it identifies real bottlenecks and memory terms rather than replacing existing domain models.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E.Synthesis/supplementary/`, `Part_E.Synthesis/outputs` and `Part_E.Synthesis/figures/`.

9 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.